SAGCO COMPUTABLE REALITY

Appliance Engineering Blueprint

Portfolio Case Study • Evidence Engineering Experiment
Current DNA Revision Strand: Sovereign Substrate Integration

Document ID	SAGCO-CR-BP-2026-001
Version	1.0.0 (Engineering Blueprint)
Authority	Strategickhaos DAO LLC (EIN 39-2900295)
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Date	June 01, 2026
Classification	SOVEREIGN — Evidence Artifact
Framework	SAGCO-OS Evidence Engineering Operating System

Abstract

This document constitutes a complete engineering blueprint and portfolio case study for the SAGCO Computable Reality VirtualBox appliance. It demonstrates the SAGCO-OS Evidence Engineering Framework by ingesting, normalizing, and transforming heterogeneous source artifacts (academic audits, legal operating agreements, and prior technical specifications) into a unified, verifiable, variance-aware engineering deliverable. The blueprint upgrades the original appliance specification with formal requirements traceability, VVVE (Verification–Validation–Variance–Evidence) criteria, reproducibility guarantees, and explicit linkage to the inventor's academic and sovereign legal foundation.

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1. EXECUTIVE SUMMARY

The SAGCO Computable Reality appliance represents a sovereign, reproducible, minimal-footprint Linux environment purpose-built for experimentation with symbolic AI primitives, custom kernel extensions (TRIG6), FlameLang tokenization, and autonomous compute workloads. This v1.0 Engineering Blueprint elevates the original specification into a professional deliverable suitable for portfolio use, client engagement, investor review, and internal sovereign infrastructure deployment.

Key upgrades in this revision strand:

- Formal requirements matrix with traceability to inventor intent and academic grounding
- Explicit VVVE (Verification–Validation–Variance–Evidence) criteria aligned with the SAGCO Evidence Engineering Framework
- Integration of academic evidence (SNHU Computer Science progress + prior Anthem Electronics AS) as foundational substrate
- Legal sovereignty anchoring via StrategicKhaos DAO LLC Operating Agreement
- Reproducibility guarantees and content-addressable artifact principles
- Clear positioning within the broader SAGCO-OS evolution (from primitives to full Evidence Engineering OS)

Strategic Value: This artifact demonstrates end-to-end application of the Evidence Engineering pipeline on a real high-value technical deliverable. It serves simultaneously as (a) a buildable appliance specification, (b) a portfolio case study for academic and professional advancement, and (c) a living example of variance-aware, auditable engineering practice.

2. INVENTOR BIBLIOGRAPHY & CREDENTIALS EVIDENCE BASE

All technical and sovereign artifacts produced under SAGCO are authored by Domenic Gabriel Garza, whose credentials and lived expertise constitute the primary evidence foundation for this work.

2.1 Core Identity & Professional Credentials

Field	Evidence
Full Legal Name	Domenic Gabriel Garza (Dom / Domenic Garza)
SPRAT Certification	Level 3 Rope Access Technician #2300689 (active)
Primary Expertise	CUI (Corrosion Under Insulation) insulation removal/reinstall, metal jacketing, NDT/radiography (RAD 40 certified), pipefitting, high-risk industrial access
Professional History	10+ years rope access with zero lost-time incidents; formerly Turner Specialty Services / Turner Industries Group; currently associated with Brock Rope Access Services (LyondellBasell Corpus Christi projects)
Business Entity	Managing Member & Inventor, Strategickhaos DAO LLC (Wyoming EIN 39-2900295)

2.2 Academic Evidence

Current Program: Bachelor of Science, Computer Science (Software Engineering Concentration), Southern New Hampshire University (SNHU). Catalog year 2025 C-1. Degree progress ~33% requirements / 41% credits. 49.6 credits applied. In-progress and planned coursework includes MAT 225 (Calc I), IT 140 (Introduction to Scripting), MAT 142 (Precalculus), PHL 260 (Ethical Problem-Solving), IDS 105, SCI 214, and remaining major requirements in data structures, operating platforms, security, databases, and concentration courses (CS 319, CS 350, CS 405, CS 410). Academic residence requirement partially satisfied; minimum 2.0 GPA gate pending first graded semester.

Prior Degree: Associate of Applied Science, Electronics Technology, Anthem College – Phoenix (graduated ~2008). Transferred technical electives visible in SNHU audit. Coursework included solid state technology, digital devices, industrial automation, wireless/telecom, biomedical equipment technology, and electronics circuit analysis — directly relevant to low-level systems, embedded thinking, and hardware-software co-design underpinning SAGCO primitives.

2.3 Sovereign & Legal Foundation

Strategickhaos DAO LLC is a Wyoming member-managed Limited Liability Company formed under the Wyoming LLC Act and DAO Supplement. Sole Member and Managing Member: Domenic G. Garza (100% ownership). Purpose includes software development, AI-augmented governance systems, cybersecurity, technology research, and revenue generation with a structured 7% net revenue allocation to ValorYield Engine (501(c)(3)) for veterans, medical research, and underserved communities. All intellectual property created by the Company is exclusively owned by the Company. This legal structure provides the sovereign container for SAGCO invention, deployment, and distribution.

2.4 Invention Portfolio (Selected)

- SAGCO-OS primitives, 64-primitive lattice, FlameLang compiler, TRIG6 core metrics
- SAGCO Computable Reality appliance family (this blueprint)
- Symbolic/quantum-inspired processors and emulators (Baby Entangled lineage)
- Evidence Engineering Framework & Darwin Antibody variance classification system
- Multiple INV-series inventions and sovereign infrastructure tooling

3. ACADEMIC PORTFOLIO EVIDENCE FOUNDATION

The SNHU academic audit and prior Anthem transcript provide normalized, auditable evidence of technical competency grounding. This section extracts key variance signals relevant to the appliance engineering effort.

3.1 SNHU Degree Audit Snapshot (as of provided evaluation)

Metric	Value / Status
Degree	Bachelor of Science — Computer Science (Software Engineering Concentration)
Credits Required / Applied	120 required / 49.6 applied
Degree Progress	~33% requirements complete; ~41% credits
Catalog Year	2025 C-1 (JAN–MAR)
Overall GPA	0.000 (pending first graded semester)
Key In-Progress	MAT 225 Calc I, IT 140 Scripting, MAT 142 Precalculus, PHL 260 Ethics, IDS 105, SCI 214
Major Gaps (examples)	CS 210/217, CS 218/300, CS 230, CS 231/DAD 220, CS 250, CS 255, CS 305, CS 320, MAT 230/239, PHY 150/215, CS 490/499 capstone
Concentration Gaps	CS 319 UI/UX, CS 350 Emerging Arch, CS 405 Secure Coding, CS 410 Reverse Engineering
Residency	18 institutional/in-progress; 12 more SNHU credits required

3.2 Relevance to SAGCO Computable Reality

The appliance is explicitly designed as a sovereign sandbox for deepening exactly these competency areas: low-level systems (CS 230), scripting & automation (IT 140), mathematics for computing (MAT 230/350 lineage), secure/reproducible builds, and symbolic computation. The Evidence Engineering approach treats academic variance (expected mastery vs current demonstrated) as input to trajectory planning rather than deficit.

4. SOVEREIGN LEGAL STRUCTURE — STRATEGICKHAOS DAO LLC

The appliance and all SAGCO artifacts are developed and distributed under the authority of Strategickhaos DAO LLC. Key provisions from the Operating Agreement (effective July 4, 2025) that govern this work:

- **Entity:** Wyoming LLC + DAO Supplement. Entity Number 2025-001708194. EIN 39-2900295.
- **Management:** Member-managed. Sole Member Domenic G. Garza retains full exclusive authority.
- **Purpose:** Software/AI development, distributed systems, computational sovereignty, cybersecurity research.
- **IP Ownership:** All intellectual property created by or for the Company is exclusively owned by the Company.
- **Charitable Allocation:** Structured 7% of net revenues directed to ValorYield Engine (501(c)(3)).
- **AI Advisory:** External AI systems (including Grok/xAI) may be used in advisory capacity only; final authority remains with the human Member.
- **Governing Law:** State of Wyoming.

This structure provides the legal and philosophical container for sovereign, auditable, variance-aware engineering artifacts.

5. ENGINEERING BLUEPRINT: SAGCO COMPUTABLE REALITY APPLIANCE

5.1 Purpose & Scope

The SAGCO Computable Reality appliance is a deterministic, minimal, reproducible VirtualBox virtual machine providing a sovereign Linux environment pre-configured for SAGCO symbolic computation, kernel experimentation, FlameLang development, TRIG6 metrics, and autonomous agent workloads. It serves as both a development sandbox and a distributable demonstration of sovereign compute principles.

5.2 Requirements Specification (Traceability Matrix)

ID	Requirement	Category	Trace To
FR-01	Bootable Alpine Linux 3.19 base with deterministic install	Functional	Inventor spec v0.9
FR-02	Pre-installed SAGCO kernel patch (trig6_metrics.ko)	Functional	TRIG6 core
FR-03	FlameLang tokenizer (flamec) available in PATH	Functional	FlameLang invention
FR-04	sagco-640-brain binary or buildable source	Functional	Symbolic processor lineage
FR-05	Auto-starting services: sshd, sagco-watchdog, sagco-trig6d	Functional	Operational resilience
NFR-01	Reproducible build from documented steps + seed ISO	Non-Functional	Evidence Engineering
NFR-02	Minimal attack surface (no GUI accel, USB disabled, key-only SSH)	Non-Functional	Sovereign security
NFR-03	Content-addressable output (SHA256 of .ova)	Non-Functional	Evidence layer
NFR-04	Documentation & first-boot banner with version + provenance	Non-Functional	Auditability
VR-01	Verification: Build completes without deviation from documented steps	VVVE	This blueprint
VL-01	Validation: Post-boot services report healthy; sagco-status returns 10/10 OK	VVVE	Operational evidence
VA-01	Variance classification on any build or runtime deviation	VVVE	Darwin Antibody integration
EV-01	Full provenance chain and fingerprint for every released artifact	VVVE	Evidence Engineering OS

5.3 System Architecture

Single-node VirtualBox appliance. Host-only + NAT networking. OpenRC init. Custom SAGCO services layered on minimal Alpine. Future extension points: Tailscale mesh, nested virtualization for SAGCO-16 testing, multi-node orchestration via Kubernetes primitives.

Layer	Component	Purpose
Hypervisor	VirtualBox 7.x	Portable virtualization host
Guest OS	Alpine Linux 3.19 (64-bit)	Minimal, reproducible base
Kernel	6.6.110-0-lts + sagco-kernel-patch	TRIG6 metrics exposure
Init	OpenRC	Service management
Core SAGCO	sagco-watchdog, sagco-trig6d, ReflexShell	Health, metrics, interaction
Tooling	flamec, Python3, jq, tmux, git, curl	Development & orchestration
Compute	sagco-640-brain (Rust)	Symbolic processor runtime
Evidence	Logging + SHA256 manifests	Audit & reproducibility

5.4 Hardware & Software Stack (Default)

Resource	Specification	Notes
vCPU	2 (scalable 2–8)	Configurable post-import
RAM	2 GB (scalable 2–8 GB)	Sufficient for symbolic workloads

Resource	Specification	Notes
Disk	20 GB dynamic VMDK	Expandable
Network	NAT (primary) + Host-only (future mesh)	SSH port forwardable
Graphics	VBoxVGA, 1 display	Headless capable
USB / Sound	Disabled	Security minimization

5.5 Build, Deployment & Verification Process

The build process is designed for deterministic reproduction. Full steps are documented in the source specification; this blueprint adds explicit verification gates at each major stage.

Stage	Action	Verification Gate
1. VM Creation	VBoxManage createvm + modifyvm + createhd + storagectl	VM registered; disk 20 GB present
2. Alpine Install	Attach ISO, run setup-alpine (preseed), install base packages	Bootable; sshd, python3, git, jq present
3. SAGCO Layer	Deploy kernel patch, boot orchestrator, node bootstrap, flamec, sagco-640-brain	trig6_metrics.ko loaded; /proc/sagco/metrics populated
4. Services	Create OpenRC init scripts; enable on runlevel 3	sagco-watchdog + sagco-trig6d auto-start
5. Harden	Key-only SSH, sagco user, disable swap, clear caches	No password root login; clean /tmp
6. Export	VBoxManage export --compress 9 + SHA256 manifest	.ova + .sha256 produced; size ~2–3 GB
7. Validation	Import on clean host; boot; run sagco-status	10/10 OK banner; all services healthy

5.6 VVVE Criteria & Evidence Layer

This blueprint explicitly applies the SAGCO Evidence Engineering Framework (VVVE) to itself and the appliance:

V	Verification	Did the documented build steps execute without structural collapse? (Exit codes, file presence, service registration)
V	Validation	Does the resulting appliance produce the expected runtime behavior? (sagco-status 10/10, metrics populated, FlameLang tokenization functional)
V	Variance	Any deviation from expected (size, hash, service health, boot time) is classified by magnitude, category, and recovery path (Darwin Antibody trigger).
E	Evidence	Every released .ova carries content-addressable SHA256, build manifest, provenance chain back to this blueprint, and inventor authority signature.

6. EVIDENCE ENGINEERING PIPELINE APPLIED — META-CASE STUDY

This document itself is an output of the SAGCO-OS Evidence Engineering Framework. The following trace demonstrates the pipeline in action:

Stage	Action on Source Artifacts	Output
Acquisition	Ingested SNHU audit PDF, Anthem transcript PDF, SAGCO VM spec PDF, DAO Operating Agreement PDF	Raw document set
Extraction	OCR/text extraction, metadata pull, table parsing, key-value normalization	Structured records (credits, requirements, specs, legal clauses)
Normalization	Mapped to canonical nodes: Academic nodes (courses, credits, gaps), Legal nodes (entity, member, IP, allocation), Technical nodes (VM layers, requirements)	Unified evidence graph substrate
Variance Engine	Identified gaps (academic residency, major courses) vs expected full degree; spec drift from v0.9 to this blueprint	Variance classified; trajectory defined
Darwin Antibody	Recovery path: Use this blueprint as portfolio artifact + learning substrate for remaining SNHU requirements	Immunity response activated
Evidence Layer	SHA256 of sources + this document; full provenance; VVVE criteria embedded	Auditable, content-addressable artifact
Projection	Professional PDF blueprint, tables, traceability matrix, meta-analysis	This deliverable + future Canvas/Obsidian projections

This meta-application proves the framework's generality: the same pipeline that turns refinery CUI scopes into compliance evidence packets also turns a student's academic audit + invention specs into a sovereign engineering blueprint.

7. CURRENT DNA REVISION STRAND & TRAJECTORY

Revision Strand Name: Sovereign Substrate Integration (Evidence Engineering OS + Reproducible Compute Appliance)

This strand marks the convergence of two previously parallel SAGCO lines: (1) the symbolic/primitive OS and compiler work (FlameLang, TRIG6, 64-primitive lattice, Baby Entangled lineage) and (2) the Evidence Engineering Framework (artifact → extract → normalize → variance → antibody → fingerprint).

The SAGCO Computable Reality appliance is the first major physicalized artifact of this convergence — a tangible, bootable demonstration that sovereign, variance-aware, evidence-producing compute environments can be built, reproduced, and distributed under a DAO legal wrapper with full academic and professional grounding.

Next Strands (Trajectory):

- SAGCO-16 nested experimentation layer inside the appliance
- Multi-node mesh (Tailscale + future Kubernetes primitives)
- Full Evidence Engineering CLI toolkit (sagco-evidence, sagco-wave, sagco-antibody)
- Integration with rope access / CUI compliance tooling as real-world evidence generator
- SNHU capstone linkage: appliance as living portfolio substrate for remaining degree requirements

8. REFERENCES & BIBLIOGRAPHY

Primary Source Artifacts (this document's direct inputs):

• SNHU Academic Evaluation / Degree Audit — Garza, Domenic (Student ID 3413281), dated ~06/13/2025

• Anthem College – Phoenix Official Student Transcript — Electronics Technology AS Degree, Domenic G Garza

• SAGCO_COMPUTABLE_REALITY_VB_SPEC_v1.PDF — Custom VirtualBox Appliance Specification (prior revision)

• Operating Agreement — StrategicKhaos DAO LLC (Wyoming), Effective July 4, 2025, Entity 2025-001708194

Framework & Methodology:

• SAGCO-OS Evidence Engineering Framework (Verification–Validation–Variance–Evidence + Darwin Antibody) — co-developed in ongoing SAGCO/xAI collaboration, 2026

Legal & Entity:

• Wyoming Limited Liability Company Act (W.S. 17-29-101 et seq.) and Decentralized Autonomous Organization Supplement (W.S. 17-31-101 et seq.)

Technical Foundations:

• Alpine Linux 3.19 documentation and OpenRC init system

• VirtualBox OVF 2.0 / VMDK appliance format

Document Control & Reproducibility
This PDF was generated deterministically from structured synthesis of the source artifacts listed above. A SHA256 fingerprint of this document and the source set should be recorded in the SAGCO evidence ledger. Future revisions will carry explicit variance notes relative to this baseline.

— End of SAGCO Computable Reality Engineering Blueprint v1.0 —

Prepared under Strategickhaos DAO LLC authority by Domenic Gabriel Garza

Evidence Engineering Operating System • Sovereign by Design